$\boxed{\text{t- test}}$ One group mean

★ Study of patients about distraction and recall of food consumed and snacking

Sample : 44 patients , 22 men and 22 women

# of biscuit intake	$\bar{X}$	S	n
Solitaire	52.1g	45.1g	22
no distraction	27.1g	26.4g	22

Condition:
① random assignment
② 22 < 10% of all distracted eaters
③ sample size / skew

(right skew)

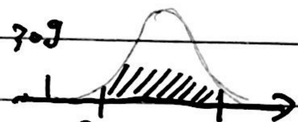
① estimate avg snack-consumption (in grams) of people who eat lunch distracted, using 95% confidence interval (two tail)

$$\underbrace{\bar{X}}_{\text{point estimate}} \pm \underbrace{t^*_{df} \cdot SE}_{\text{margin of error}} = 52.1 \pm t^*_{21} \cdot \frac{S}{\sqrt{n}}$$

$$= 52.1 \pm 2.08 \cdot \underbrace{\frac{45.1}{\sqrt{22}}}_{\gg 9.62}$$

$$= 52.1 \pm 20$$

$$= (32.1,\ 72.1)$$

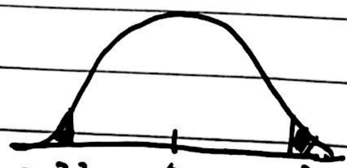


② Suppose the suggested size of these biscuit is 30g. Does the data provide convincing evidence the amount of snacks consumed by distracted eaters post-lunch is different than the suggested size (30g)?

$H_0: \mu = 30g$
$H_A: \mu \neq 30g$

$$T = \frac{\bar{X} - \mu}{SE} = \frac{52.1 - 30}{9.62}$$

$$= 2.3$$



$p \approx 0.0318$ → Reject H_0 → 95% confidence interval doesn't aggree too.

t-test-2

Comparing two independent means

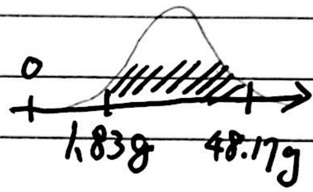
biscuit intake	$\bar{X}$	s	n
Solitaire	52.1g	45.1g	22
no distraction	27.1g	26.4g	22

$$SE = \sqrt{\frac{S_1^2}{N_1} + \frac{S_2^2}{N_2}}$$

① Estimate the difference between the average post-meal snack consumption between those who eat with and without distractions.

point estimate ± margin error

$$(\bar{X}_d - \bar{X}_{nd}) \pm t^*_{df} \cdot SE$$



$$= (52.1 - 27.1) \pm 2.08 \times \underbrace{\sqrt{\frac{45.1^2}{22} + \frac{26.4^2}{22}}}_{11.14}$$

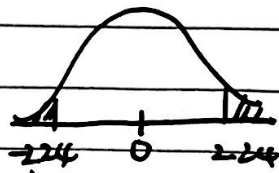
$$= 25 \pm 23.17 = (1.83g,\ 48.17g)$$

② Do these data provide convincing evidence of a difference between solitaire group and no distraction groups?

H_0: $\mu_d - \mu_{nd} = 0$

H_A: $\mu_d - \mu_{nd} \neq 0$

$$T_{df=21} = \frac{25-0}{SE} = \frac{(\bar{X}_d - \bar{X}_{nd}) - 0}{11.14} = 2.24$$



p-value ≈ 0.04 → reject H_0

* from class: Since we have a randomized control trial, if we do find a significant result, we could talk about a causal relationship between two variables.

Comparing two paired means

* student's reading and writing scores. (200 students)

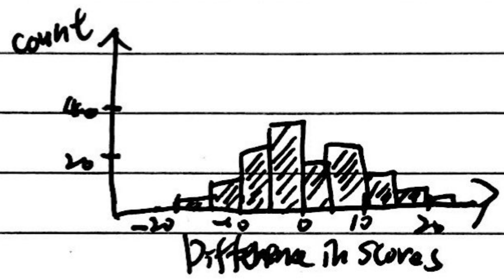
Student ID	read	write	diff
70	57	52	5
86	44	33	11
141	63	44	19
172	47	52	-5
⋮	⋮	⋮	⋮
137	63	65	-2

parameter of interest	point estimate
Average difference of all students.	Average difference of sampled students
μ_{diff}	$\bar{x}_{diff}$

$\bar{x}_{diff} = -0.545$

$s_{diff} = 8.887$

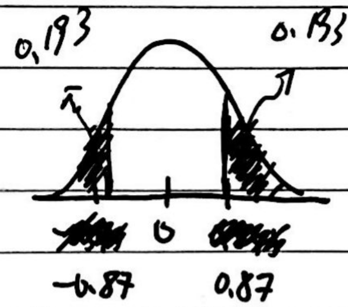
$n_{diff} = 200$



$H_0 : \mu_{diff} = 0$

$H_A : \mu_{diff} \neq 0$

$$T = \frac{(-0.545) - 0}{\frac{8.887}{\sqrt{200}}} = -0.87$$



$df = 200 - 1 = 199$

p-value = 0.386

p-value: probability of obtaining a random of 200 students where the average difference in scores is at least 0.545, given the truth average difference in scores is 0.

p-value : P(observed or more extreme outcome (from data) | H_0 is true)